ELG 3125 - Signals & Systems

Lecture 6 - Part 1

Differential & Difference Equations

A class of systems for which the input & output are related through:

- A linear constant-coefficient differential equation in continuous time, or
- A linear constant-coefficient difference equation in discrete-time.

Linear Constant-Coefficient Differential Equation

An equation that has a function and its derivative in it.

Ex: $\frac{dy(t)}{dt} + 2y(t) = x(t)$, $y(t)$: output, $x(t)$: input

N-th order linear constant coefficient differential equation: $\boxed{\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}}$

Solution to the differential equation: $y(t) = \underbrace{y_h(t)}_{\text{homogenous}} + \overbrace{y_p(t)}^{\text{particular}}$

Example: Solve the system: $\frac{dy(t)}{dt} + 2y(t) = x(t)$. $x(t) = ke^{3t}u(t)$, $k \in \mathbb{R}$

$y_h(t) = ?$

Guess: $y_h(t) = Ae^{\lambda t}$, $\frac{dy_h(t)}{dt} = \lambda Ae^{\lambda t}$

$\frac{dy_h(t)}{dt} + 2y_h(t) = 0 \Rightarrow \lambda Ae^{\lambda t} + 2Ae^{\lambda t} = 0 \Rightarrow \underbrace{Ae^{\lambda t}}_{\neq 0}(\lambda + 2) = 0 \Rightarrow \boxed{\lambda = -2}$

$\boxed{y_h(t) = Ae^{-2t}}$ is a solution to the D.E $\forall A$.

$y_p(t)$ has the same form as $x(t)$

$\Rightarrow y_p(t) = Ye^{3t}$, $x(t) = ke^{3t}u(t) = \begin{cases} ke^{3t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$, $\frac{dy_p(t)}{dt} = 3Ye^{3t}$

$\Rightarrow \frac{dy_p(t)}{dt} + 2y_p(t) = x(t) \Rightarrow 3Ye^{3t} + 2Ye^{3t} = ke^{3t} \Rightarrow 3Y + 2Y = k \Rightarrow 5Y = k \Rightarrow \boxed{Y = \frac{k}{5}}$

$\boxed{y_p(t) = \frac{k}{5}e^{3t}, \; t > 0}$

$\boxed{y(t) = y_h(t) + y_p(t) = Ae^{-2t} + \frac{k}{5}e^{3t}, \; t > 0}$

↓ initial conditions required to find A.

But for LTI systems: $y(0) = 0$

$\Rightarrow y(0) = Ae^{-2(0)} + \frac{k}{5}e^{3(0)} = A + \frac{k}{5} = 0 \Rightarrow \boxed{A = -\frac{k}{5}} \Rightarrow \boxed{y(t) = \frac{k}{5}(e^{3t} - e^{-2t}), \; t > 0}$

Linear Constant-Coefficient Difference Equations

N-th order linear constant coefficient difference equation has the form: $\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$

$y[n] = y_h[n] + y_p[n]$

Concept of initial rest of the LTI causal system described by the difference equation means that $x[n]=0, n<n_0$ implies $y[n]=0, n<n_0$.

Example: Consider the difference equation, $x[n]=k\delta[n]$

$$y[n] - \frac{1}{2}y[n-1] = x[n]$$

$$y[n] = \frac{1}{2}y[n-1] + x[n] = \frac{1}{2}y[n-1] + k\delta[n]$$

$$\Rightarrow y[n] = \frac{1}{2}y[n-1] + k\delta[n]$$

$$y[0] = \frac{1}{2}\overbrace{y[-1]}^{=0 \text{ (LTI condition)}} + k\underbrace{\delta[0]}_{=1} = k$$

$$y[1] = \frac{1}{2}y[1-1] + k\cancel{\delta[1]}^{\,0} = \frac{1}{2}y[0] = \frac{1}{2}k$$

$$y[2] = \frac{1}{2}y[2-1] + k\cancel{\delta[2]}^{\,0} = \frac{1}{2}y[1] = \frac{1}{2}\left(\frac{1}{2}\right)k = \left(\frac{1}{2}\right)^2 k$$

$$y[3] = \frac{1}{2}y[3-1] + k\cancel{\delta[3]}^{\,0} = \frac{1}{2}y[2] = \frac{1}{2}\left(\frac{1}{2}\right)^2 k = \left(\frac{1}{2}\right)^3 k$$

$$\vdots$$

$$y[n] = \frac{1}{2}y[n-1] + k\delta[n] = \left(\frac{1}{2}\right)^n k \Rightarrow \boxed{y[n] = k\left(\frac{1}{2}\right)^n u[n]}$$